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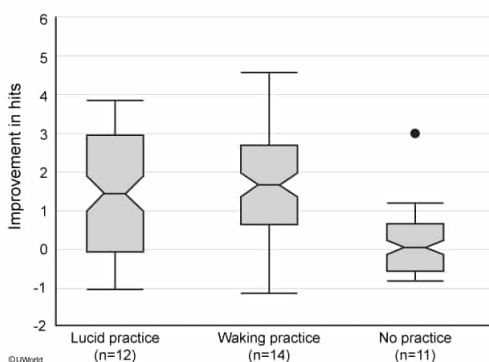


Figure 1 Median improvement in total hits for each group (Note: The notches represent the 95% confidence interval for the median.)

The researchers proposed that a follow-up study be devised and conducted to control for any potential variability in each participant's sleep at home. The researchers suggested using the same study procedure except participants would sleep overnight for five consecutive nights in a sleep lab, where the timing and duration of sleep would be standardized. In order to eliminate any potential "first-night effects" (an initial reduction in sleep quality when sleeping in an unfamiliar environment), researchers proposed that pretest and posttest measures each be taken before and after the third, fourth, and fifth night's sleep.

A mechanism for researchers to monitor a dream state would be provided by tracking which of the following?

- ✓ ☒ A. Closed-eye movements [79%]
- ☐ B. Electroencephalography measurements of K-complexes [17%]
- ☐ C. Spikes in growth hormone release [1%]
- ☐ D. Sudden body movements [1%]

Explanation:

	Awake	→ Stage 1	→ Stage 2	→ Stages 3 & 4	→ REM
Characteristics		• Fleeting thoughts • Hallucinations • Muscle twitches	• Increased relaxation	• Deepest sleep • Growth hormone (GH) released	• Muscle paralysis • Dreaming
Heart rate	100 bpm				
Respiration	12 - 16 breaths/min.				
Blood pressure	120/80				
Body temperature	98.6°F				

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Dreaming is most often associated with **REM sleep**, particularly the visually intense dreams remembered after waking. Also known as paradoxical sleep, REM sleep is characterized by **brain activity** (measured by electroencephalography [EEG]) that **resembles** the beta waves seen in an **awake, alert state**, except there is **no muscle tone** in the body. This prevents the dreamer from acting out the dream, which could result in injury. The only exceptions to this systemic paralysis are the muscles controlling the eyes and the cardiopulmonary system.

Although the body is functionally paralyzed during REM sleep, the eye muscles are very active and cause the characteristic closed-eye movements (ie, **rapid eye movements**) from which this state derives its name. Therefore, researchers can track closed-eye movements to monitor a dream state.

Physiological changes are also characteristic of REM sleep. Cardiovascular and respiratory activities become very regular and gradually decrease, along with temperature, during NREM sleep. During REM sleep, **body temperature drops** to its lowest point, while **cardiovascular** and **respiratory** activities become **irregular** and **increase**.

(Choice B) EEG measurements showing **K-complexes** demonstrate the presence of stage 2 sleep.

(Choice C) Spikes in the release of growth hormone (GH) coincide with deep sleep stages 3 and 4.

(Choice D) Sudden, jerking body movements occur when an individual first falls asleep and enters stage 1 sleep.

Educational objective:

During NREM sleep, cardiopulmonary activity becomes regular and decreases, along with body temperature. During REM sleep, cardiopulmonary activity becomes irregular and increases, while body temperature drops to its lowest point. REM sleep is also characterized by rapid closed-eye movements, muscle paralysis, and brain activity resembling a waking state.

Time spent: 0 sec
Subject:
 Behavioral Sciences

QID: 400137
System:
 Sensation, Perception, and Consciousness

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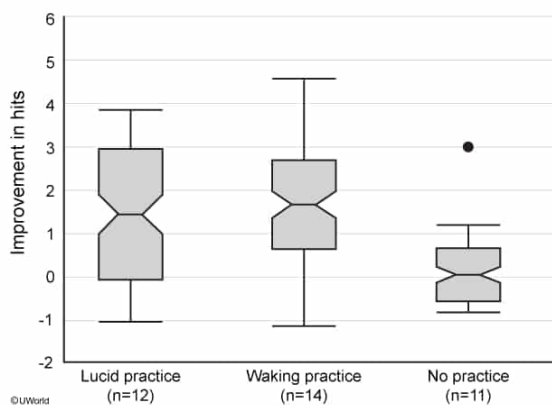


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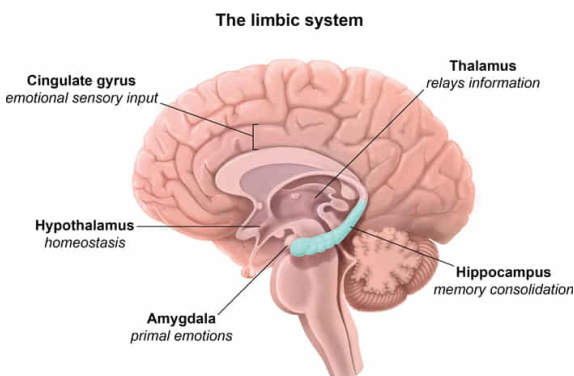
the following groups, if added to the current study design, would best accomplish this goal?

- ☐ A. A group of lucid dreamers who practiced the task both while sleeping and while awake. [11%]
- ✓ ☒ B. A group of nonlucid dreamers who practiced the task but did not sleep between the pretest and posttest. [63%]
- ☐ C. A group of nonlucid dreamers who did not practice the task and did not sleep between the pretest and posttest. [9%]
- ☐ D. No additional groups are needed; this can be accomplished by comparing the results of the "waking practice" group with the "no practice" group. [14%]

Correct

63% answered correctly

Explanation:



There is growing evidence that sleep is a vital part of **memory consolidation**, a series of processes that stabilizes a memory trace after its initial acquisition during waking consciousness. The **hippocampus**, which is part of the limbic system, plays an integral role in memory consolidation. The **limbic system**, a network of brain regions that also includes the amygdala, hypothalamus, thalamus, and cingulate gyrus, is involved in **emotion, learning, and memory**.

In order for the researchers to examine the role of sleep in memory consolidation, the best group to add to the current study design would be a group of nonlucid dreamers who practiced the task but *did not sleep* between the pretest and posttest ("waking practice/no sleep").

By comparing the results of the "waking practice/no sleep" group with those of the "waking practice" group, the researchers could examine if sleep was helpful for consolidating memory generated through task practice because the two groups differ by only one variable, the presence or absence of sleep.

(Choice A) Adding a group of lucid dreamers who practiced the task while sleeping and while awake would not be useful because there would be no comparison group that did not

sleep, and it would not be possible to differentiate the effects of waking practice vs. sleeping practice within this group.

(Choice C) Adding a "no practice/no sleep" group would be less useful than adding a "waking practice/no sleep" group. Researchers could compare the results of the "no practice/no sleep" group with the "no practice" group to see if no practice alone produces little task improvement (the two groups would have similar results) or if no practice and no sleep is worse than no practice alone. Because the added group would not be practicing, however, it would be less useful for examining if sleep plays a role in *memory consolidation*.

(Choice D) Since both the "waking practice" and "no practice" groups slept, it is not possible to draw any conclusions about the role of sleep in memory consolidation by comparing the results from these two groups.

Educational objective:

The hippocampus plays a critical role in the formation of new memories through the process of memory consolidation, which stabilizes a memory trace after its initial acquisition during waking consciousness. Sleep appears to play an important role in memory consolidation.

Time spent:0 sec

Subject:
Behavioral Sciences

QID:400138

System:
Sensation, Perception, and Consciousness

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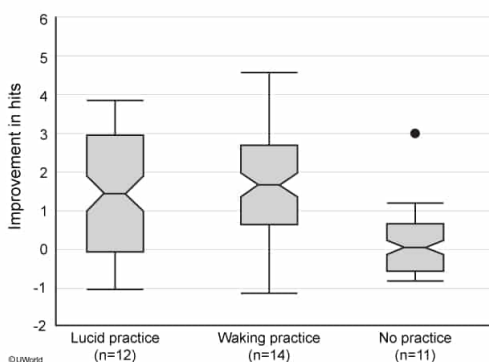


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Compared with the study described in the second paragraph, the proposed follow-up study should demonstrate:

- ☐ A. less internal and external validity.
- ☐ B. less internal validity but greater external validity. [9%]
- ☒ C. greater internal validity but less external validity. [49%]
- ☐ D. greater internal and external validity. [40%]

Explanation:

	Internal validity	External validity
	Causality	Generalizability
Describes:	<i>If a change in the independent variable is really causing a change in the dependent variable</i>	<i>If the observed relationship applies outside the experiment to another situation or other people</i>
Major threats to:	• Confounding variables • Selection bias • Maturation • Repeated testing • Regression toward the mean	• Artificial research environment • Nonrepresentative sample • Measurement effects
As experiment becomes more tightly controlled:	↑	↓
As experiment becomes more like the real world:	↓	↑

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In general, as a study becomes more tightly controlled and confounding variables are limited, internal validity tends to increase but external validity tends to decrease.

Internal validity is mostly concerned with **causality**, or the extent to which changes in the dependent variable can be attributed to changes in the independent variable. The presence of **confounding variables**, which are additional variables that might influence results or outcomes, **decreases internal validity** by introducing the possibility that results are attributable to a confounding variable and not to the independent variable of interest.

External validity is mostly concerned with **generalizability**, or the extent to which results can be **applied to other situations** (eg, outside the laboratory) or beyond the sample to the larger population.

Allowing participants a single night's sleep at home introduces many variables not controlled by the study, such as time of sleep onset, quality of sleep, and duration of sleep. Standardizing the timing and duration of sleep by having participants sleep in a sleep lab creates an unnatural sleeping environment for the subjects. Therefore, the proposed follow-up study should demonstrate greater internal validity, because there should be fewer confounding variables, but less external validity, because the findings will be less applicable to a population with a more natural sleep experience.

(Choices A and B) If the researchers repeat the study and better control for sleep onset and duration, this will result in *greater* (not less) internal validity.

(Choice D) Limiting confounding variables occurring in real life *decreases* the external validity of a study.

Educational objective:

Internal validity measures the extent to which a causal relationship can be concluded from the study, and external validity measures the extent to which the study results can be generalized beyond the study. Typically, increases in internal validity result in decreases in external validity, and vice versa.

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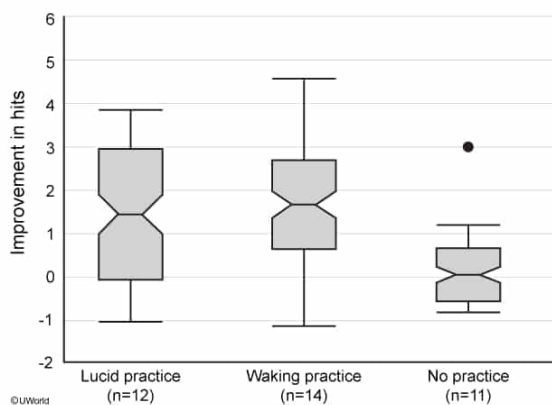


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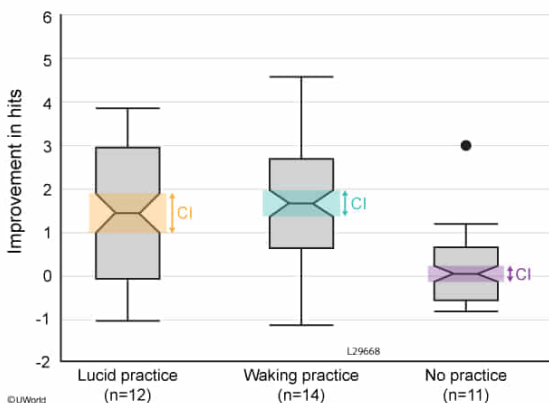
Which of the following conclusions is most supported by Figure 1?

- ☐ A. An individual from the "lucid practice" group demonstrated the lowest improvement of all members. [1%]
- ✓ ☒ B. The difference in task improvement between the "lucid practice" group and the "no practice" group was statistically significant. [46%]
- ☐ C. The "lucid practice" group's lowest quartile improved more on the task than the "no practice" group's highest quartile. [25%]
- ☐ D. The mean score for the "waking practice" group was higher than the mode scores for the other two groups. [26%]

Correct

45% answered correctly

Explanation:



The results of the lucid dreaming study described in the passage are illustrated with a **box-and-whisker plot**, which is used to visually represent the **median** and the **distribution** of the data into **quartile ranges**:

- The **upper quartile** (Q3; 75% of the data lies below this value), shown as the top edge of the box.
- The **median** (50% of the data lies below this value), which is the line inside the box.
- The **lower quartile** (Q1; 25% of the data lies below this value), shown as the bottom edge of the box.
- The **maximum** and **minimum data points**, shown as whiskers, are data values no more than 1.5 times the interquartile range (IQR) from the edge of the box. The IQR is equal to the difference between the upper and lower quartiles.
- **Outliers**, shown as data points outside the maximum and minimum data points.

Confidence intervals (CIs) represent a measure of uncertainty in a reported value by indicating how far the value reported might be from the *true* value (eg, true median). Theoretically, a 95% CI indicates that if the same population were sampled countless times and CIs were calculated each time, the resulting intervals would contain the *true* population value in approximately 95% of the cases.

A notch that represents the 95% CI of the median is used on the box-and-whisker plot to estimate visually whether the medians of two groups are significantly different. If the

notches across two groups **do not overlap**, the **medians** of the two groups are probably **significantly different**.

In Figure 1, the notches of both the "waking practice" and "lucid practice" groups are higher than and do not overlap with the notch of the "no practice" group, so the two practice groups are assumed to have improved significantly more on the task than the "no practice" group. However, one cannot conclude from the figure that a significant difference exists between groups when notches do overlap.

(Choice A) An individual from the "waking practice" group (not the "lucid practice" group) demonstrated the lowest improvement of all members.

(Choice C) The "lucid practice" group's lowest quartile was completely below the "no practice" group's highest quartile, indicating less improvement for the "lucid practice" group's lowest quartile.

(Choice D) Box-and-whisker plots provide information about the median, not the mean or the mode.

Educational objective:

Box-and-whisker plots visually represent the median value and interquartile range. Notched box plots show the 95% confidence interval around the median. If notches do not overlap, medians can generally be inferred to be statistically significantly different.

Time spent:0 sec

Subject:

Behavioral Sciences

QID:400140

System:

Sensation, Perception, and Consciousness